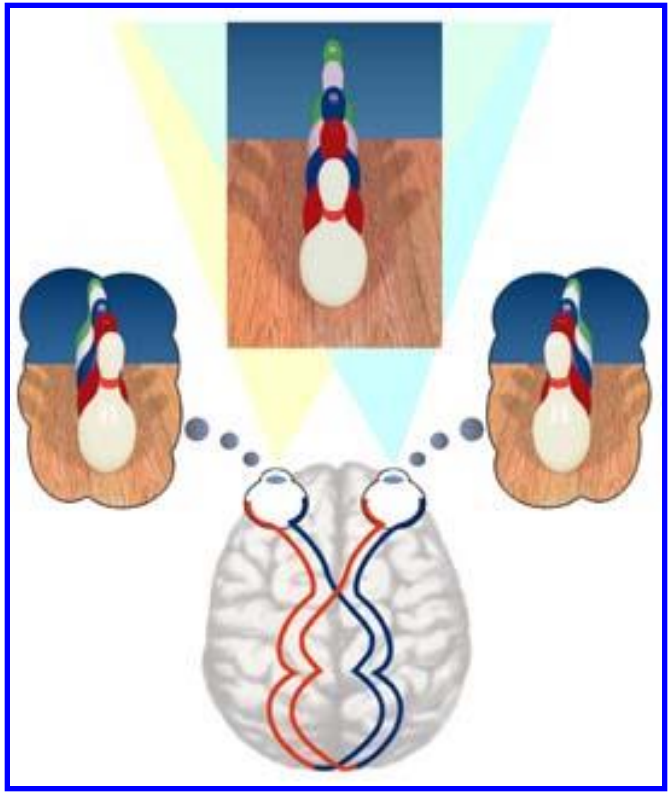
Depth from Stereo

Avinash Sharma

CSE Department, IIT Jodhpur

Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Stereo Imaging



1. We see a slightly different image of the world through the two eyes.

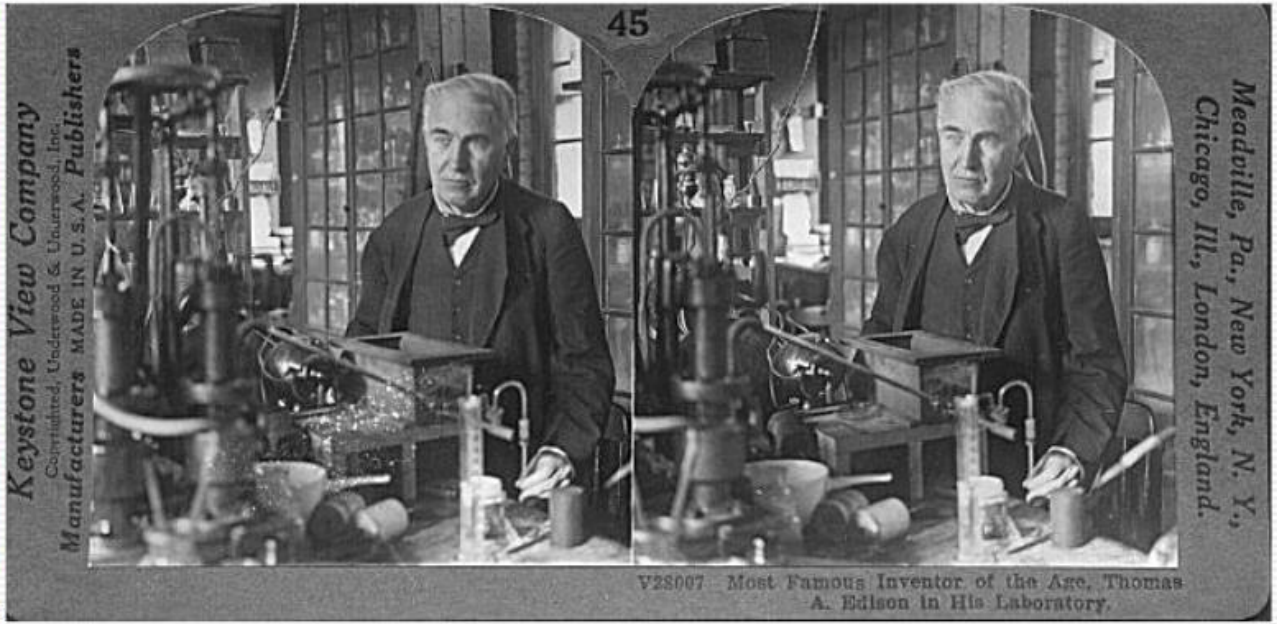
2. The shift in image is proportional to the the distance to the object.

3. The roadside trees are left behind, while the farther mountains follow you.

http://www.vision3d.com/stereo.html

Stereo Imaging

© Copyright 2001 Johnson-Shaw Stereoscopic Museum

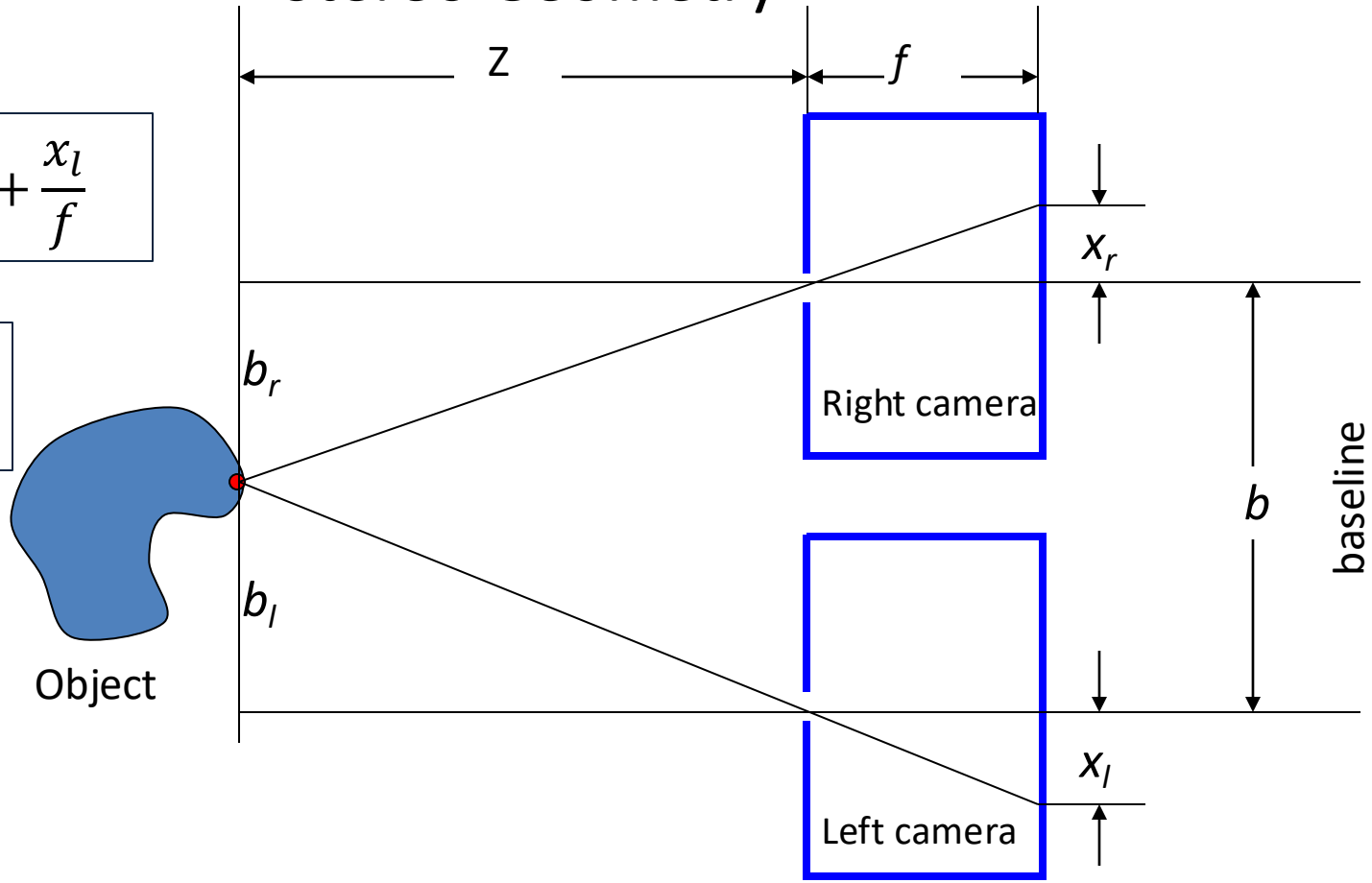


Slide source: K. Grauman.

Stereo Geometry

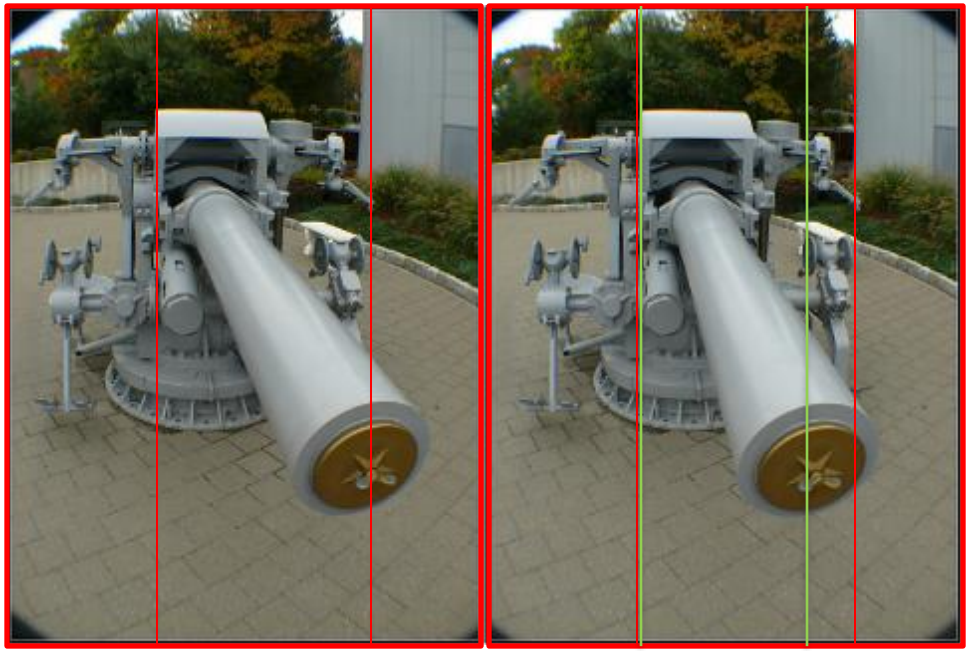
$$\frac{b_r}{Z} + \frac{b_l}{Z} = \frac{x_r}{f} + \frac{x_l}{f}$$

$$Z = \frac{f.b}{(x_r + x_l)}$$



Stereo Geometry

- Farther the point, smaller the disparity and vice versa.
- A large baseline can give more reliable estimates of depth. But, matching may become harder
- Basic step: Identify common points in camera views
  - How do we find the images of a single world point in two views?
  - Do they have the same intensity/color?
  - What else can be same/similar?



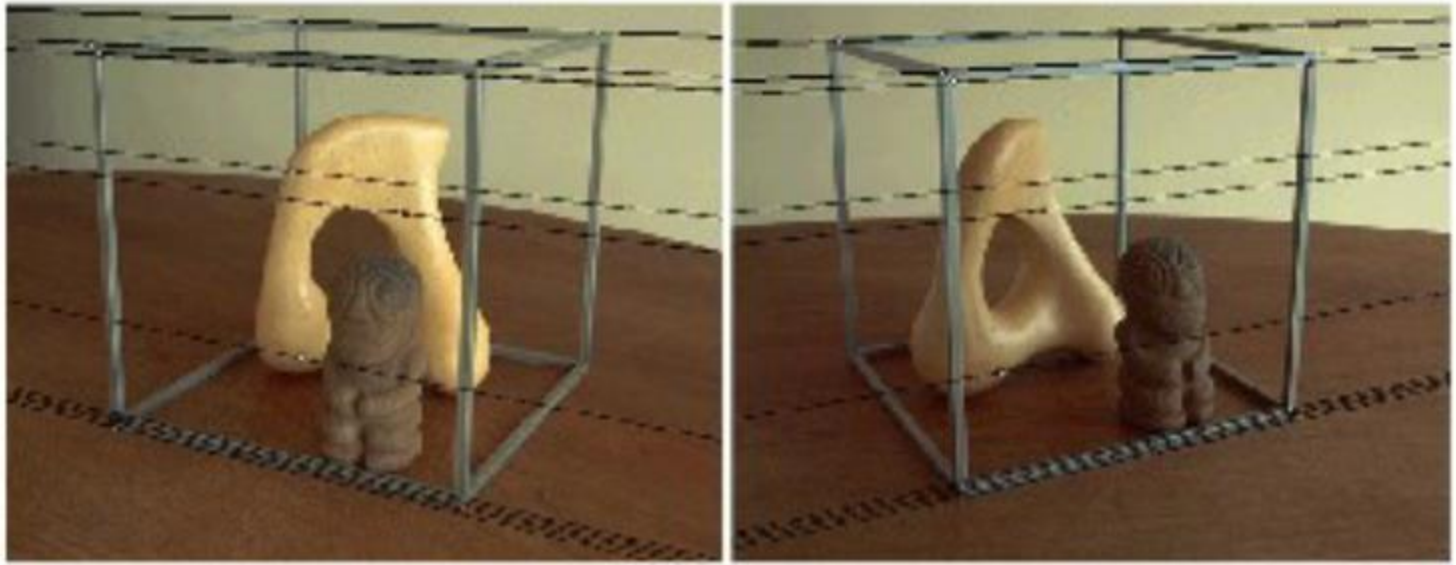
Identifying Common Points

- Find a world point in 2 or more views
- Appearance is the only clue to identify them
- Individual pixel colors are similar very often. However, the match is too noisy.
- Match a (small) neighborhood of colors from one image to a similar neighborhood in others
- Will work if local surface is fronto-parallel and images have similar magnification
- Foreshortening can happen when viewing an oblique surface
- Many ambiguities. We need a lot of help!

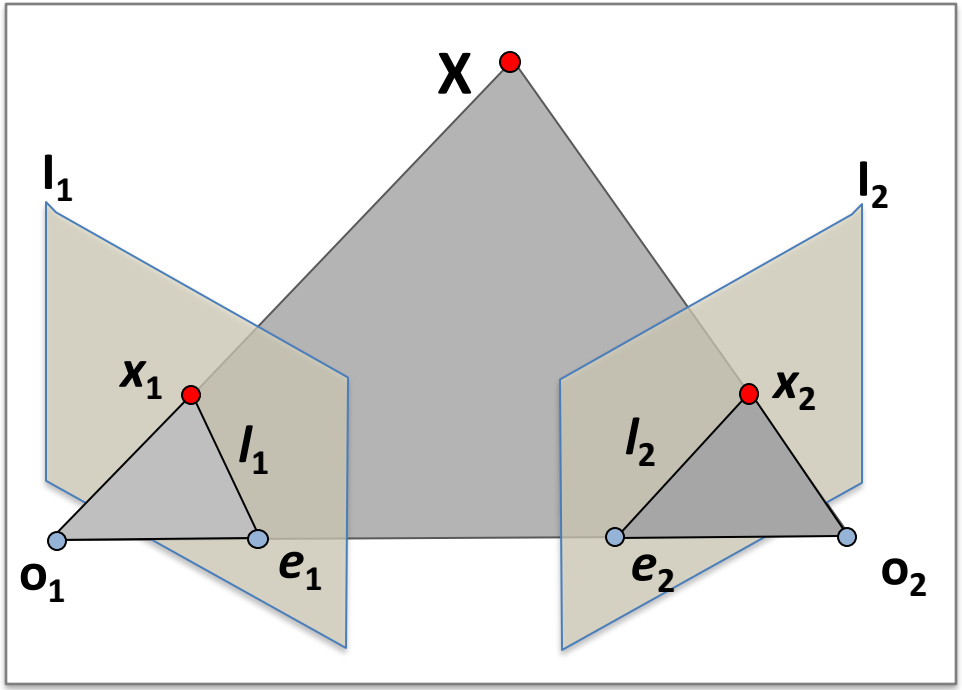
Some Examples

Constraints on Matching

- **Epipolar:** Match lies on the epipolar line of the pixel

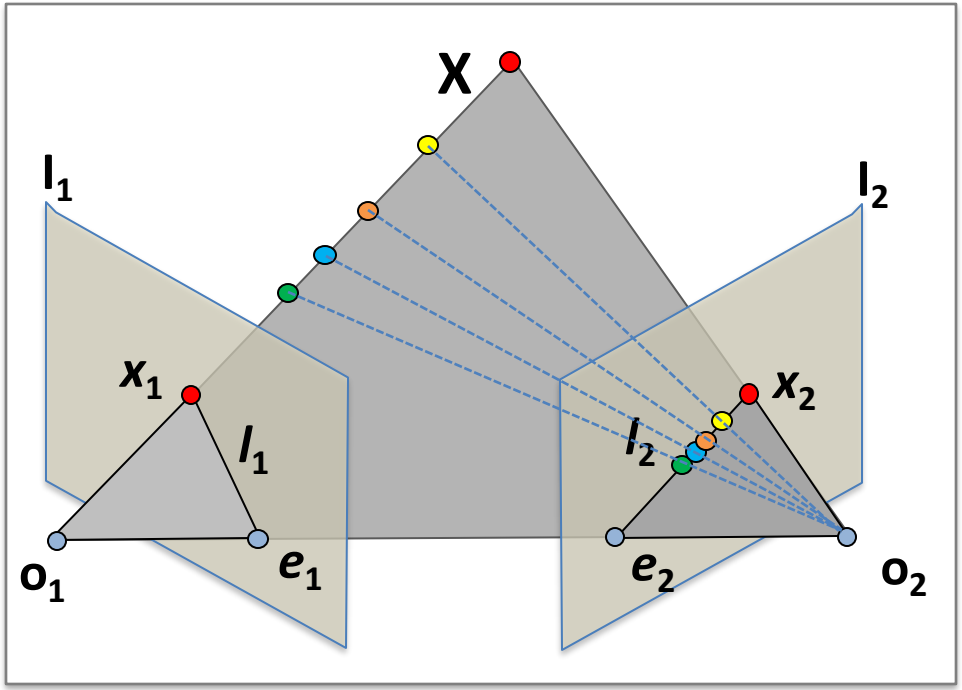


Epipolar Geometry



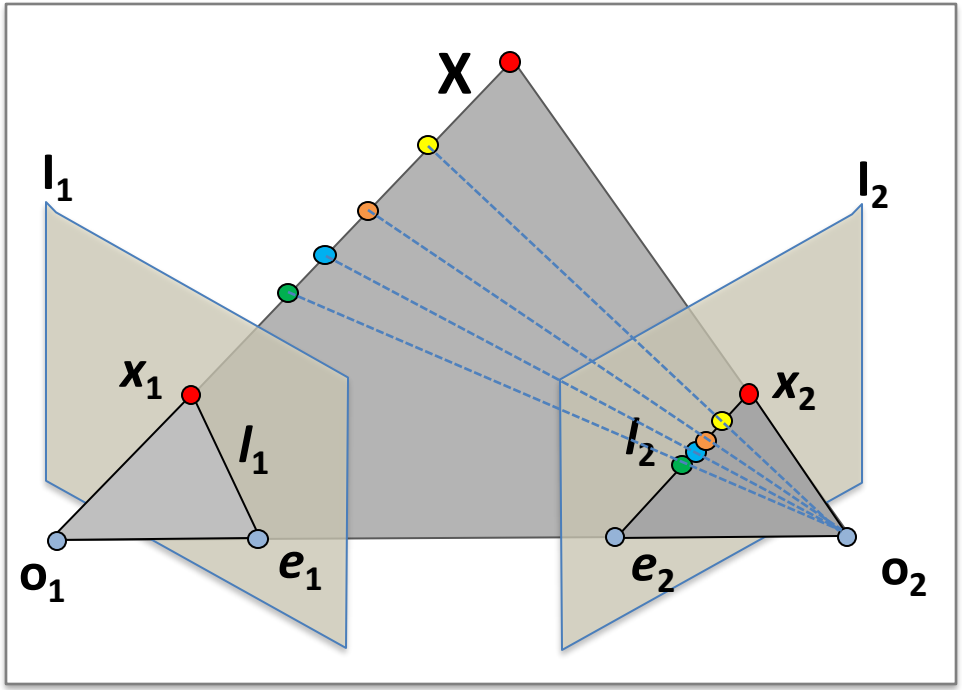
- All world points that map to x_1 in I_1 (pre-image of x_1) map to a line l_2 in I_2, called an epipolar line. (vice-versa)
- The image of o_1 in I_2 (e_2) is an epipole. (vice-versa)

Epipolar Geometry



- All world points that map to x_1 in I_1 (pre-image of x_1) map to a line l_2 in I_2, called an epipolar line. (vice-versa)
- The image of o_1 in I_2 (e_2) is an epipole. (vice-versa)

Epipolar Geometry



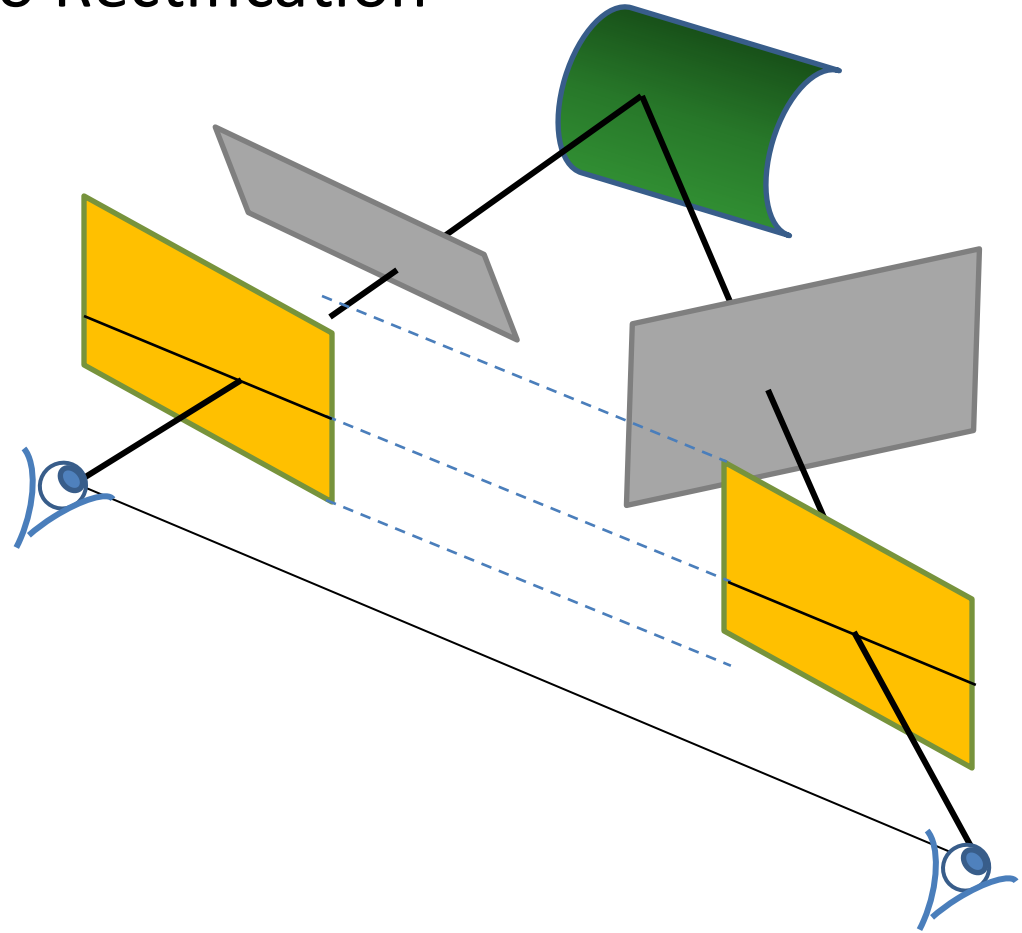
- The plane containing these is called the epipolar plane.
- These result in a set of constraints, which are referred to as the *epipolar constraints* and the resulting geometry is called the *epipolar geometry.*

Constraints on Matching

- **Epipolar:** Match lies on the *Epipolar Line* of the pixel
- **Colour Constancy:** The appearance/colour does not change from one view to another
- **Uniqueness:** A point on left image can match with only one point on the right and vice versa
- **Ordering or Monotonicity:** If point A is to the left of B in view 1, it will to the left of B in view 2 also
- **Continuity:** Disparity values vary smoothly (violated at occlusion boundaries)
- **Sparse correspondence:** only for good feature points
- **Dense correspondence:** a match for every pixel

Stereo Rectification

- Reproject images onto a common plane parallel to the line between the optical centers
- Pixel motion is horizontal after this transformation



C. Loop and Z. Zhang "Computing Rectifying Homographies for Stereo Vision", IEEE Computer Vision and Pattern Recognition 1999

Stereo Rectification

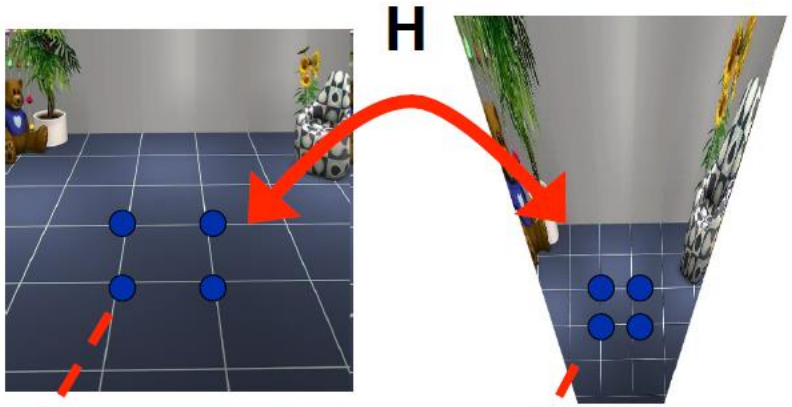
1. Estimate **E** using the 8 point algorithm (SVD)
2. Estimate the epipole **e**
3. Build $\mathbf{R}_{rect}$ from **e**
4. Decompose **E** into **R** and **T**
5. Set $\mathbf{R}_1 = \mathbf{R}_{rect}$ and $\mathbf{R}_2 = \mathbf{R}\mathbf{R}_{rect}$
6. Rotate each left camera point (warp image) $[x' \; y' \; z'] = \mathbf{R}_1 [x \; y \; z]$
7. Rectified points
8. Repeat 6 and 7 for right camera points using $\mathbf{R}_2$

Stereo Rectification

Estimating Epipoles from Essential Matrix

1. Obtain the Essential Matrix E .
2. Perform SVD: on E: $E = U \Sigma V^T$.
3. Identify the zero singular value: in the Σ matrix.
4. Extract the corresponding columns: from U and V.
    4. The column of V corresponding to the zero singular value is the epipole e_0.
    5. The column of U corresponding to the zero singular value is the epipole e_1.

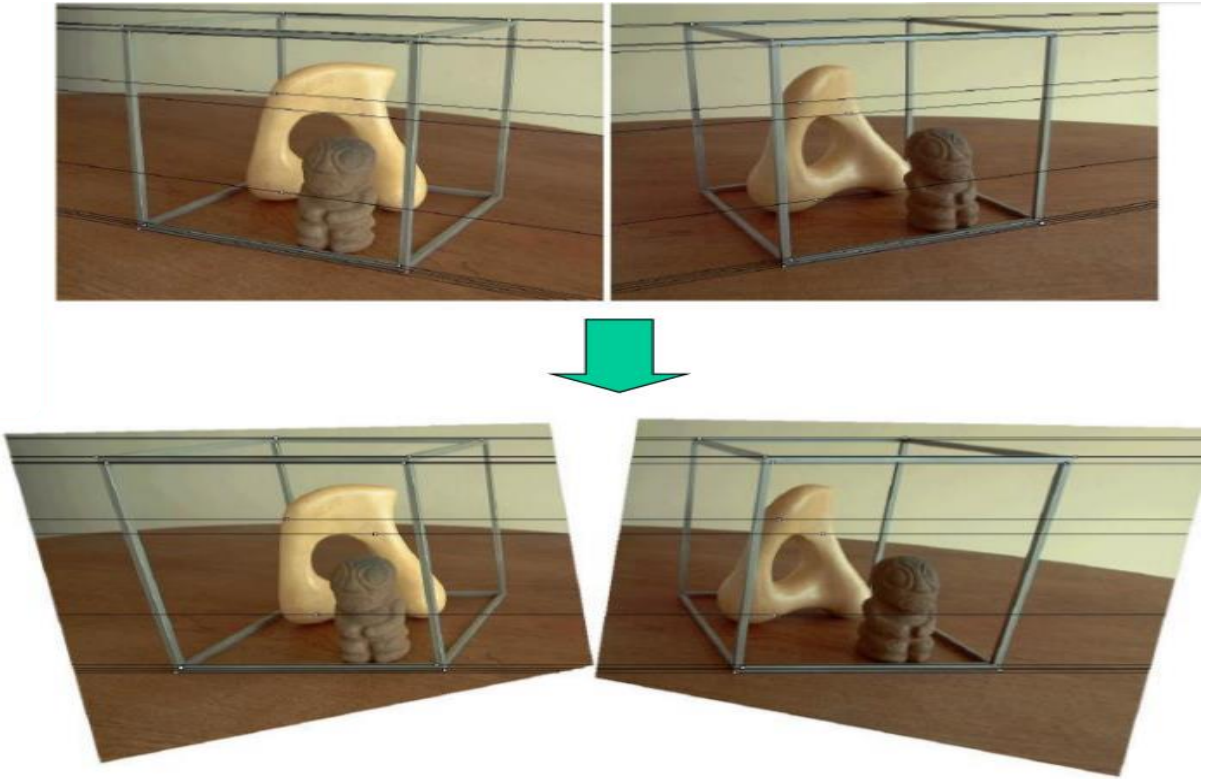
Stereo Rectification Example



x_i $\quad$ x'_i

$$x'_i = H\, x_i$$

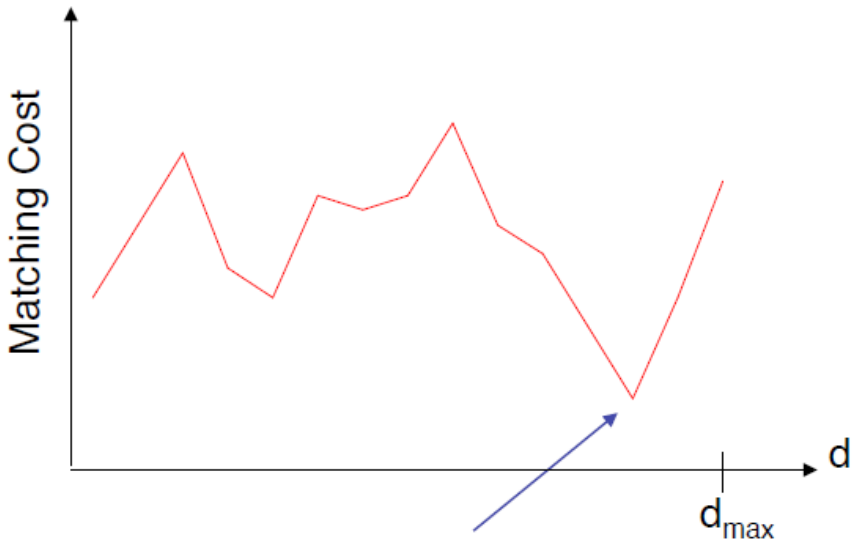
Stereo Rectification Example



Matching Pixels

- Disparity Computation (search over scanlines)
  - Similarity of pixel based on intensity (absolute or squared) difference
    - AD : $|I_L(x,y) - I_R(x,y)|$, SD: $(I_L(x,y) - I_R(x,y))^2$

$$|I_L(x,y) - I_R(x+d,y)|$$

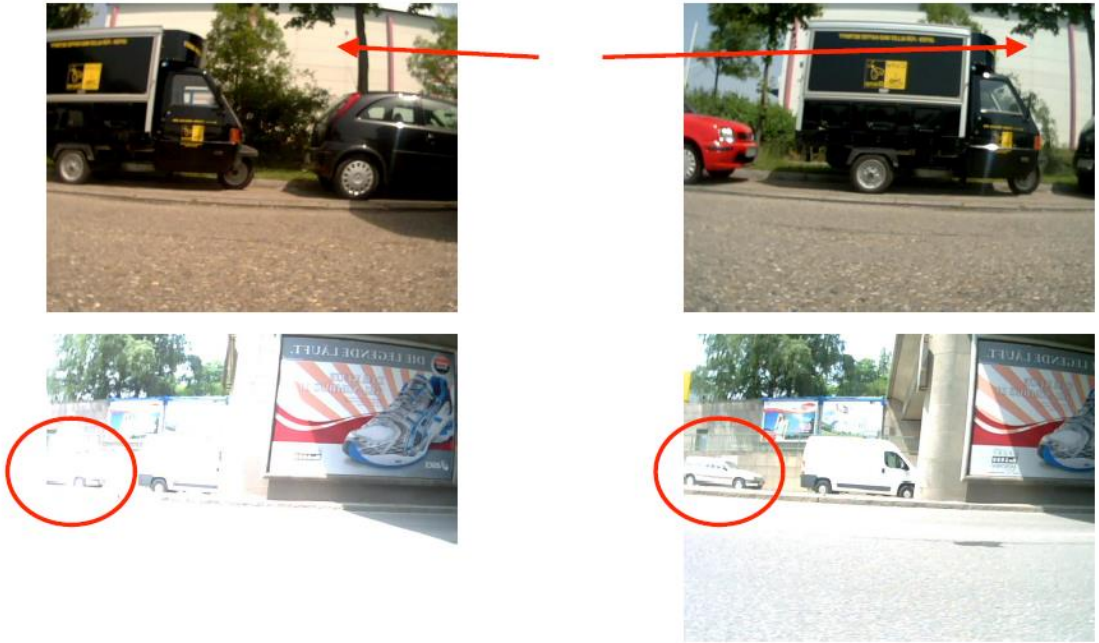


Matching Patches v/s Pixels

- Low signal-to-noise ratio causes noisy disparity map

Challenges

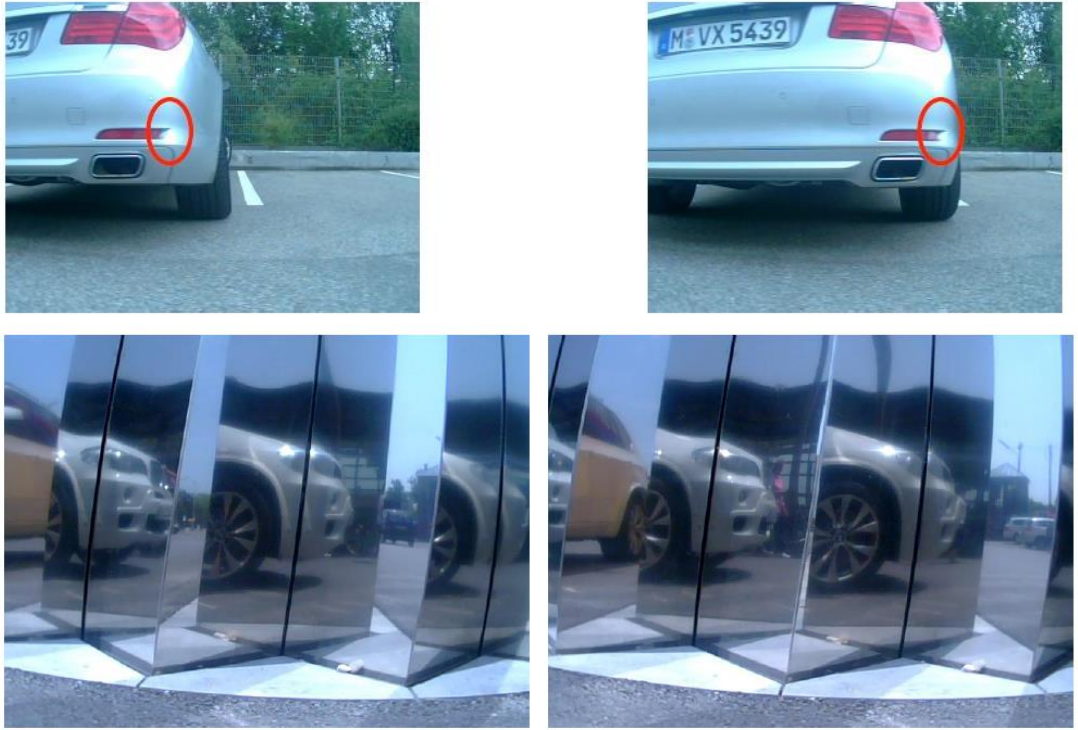
- Photometric Variations/Sensor Noise



Source C. Unger

Challenges

- Specularities / Reflections



Source C. Unger

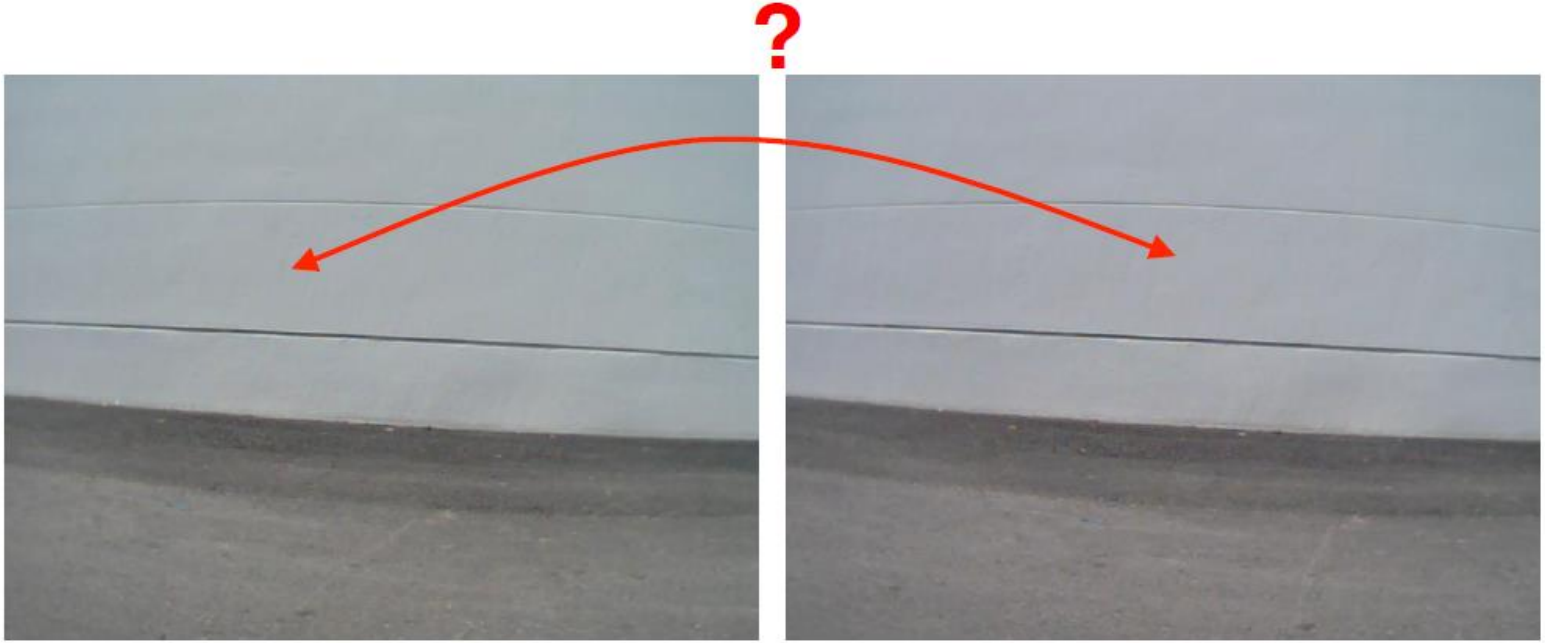
Challenges

- Foreshortening & Uniqueness

Source C. Unger

Challenges

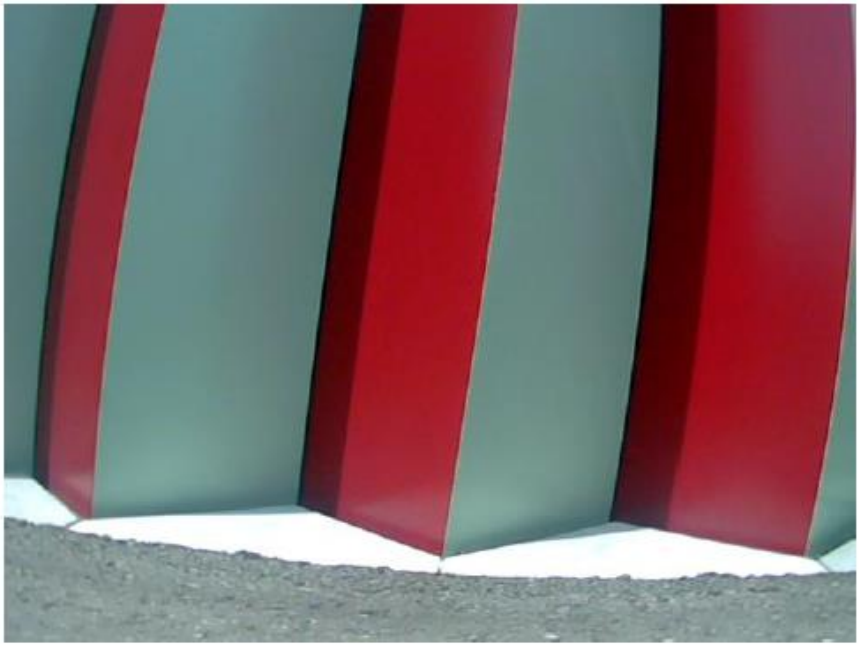
- Textureless Regions



Source C. Unger

Challenges

- Repetative Textures/Patterns



Source C. Unger

Challenges

- Occlusions

Source C. Unger

Challenges

- Transparent/Translucent Surfaces

Source C. Unger

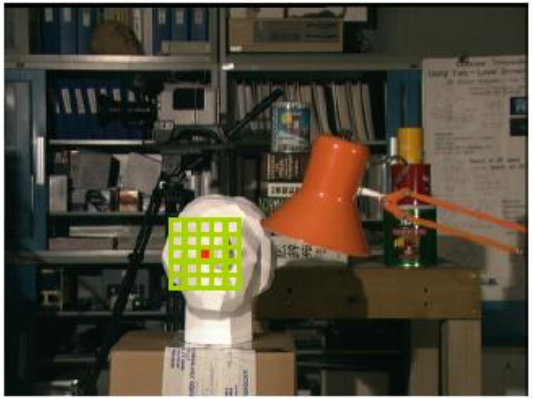
Challenges

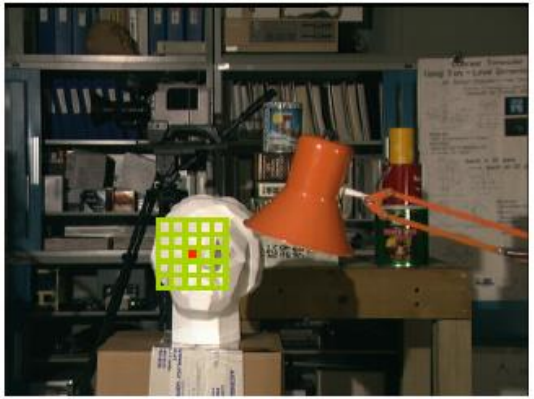
- Perspective Distortion

Source C. Unger

Matching Patches

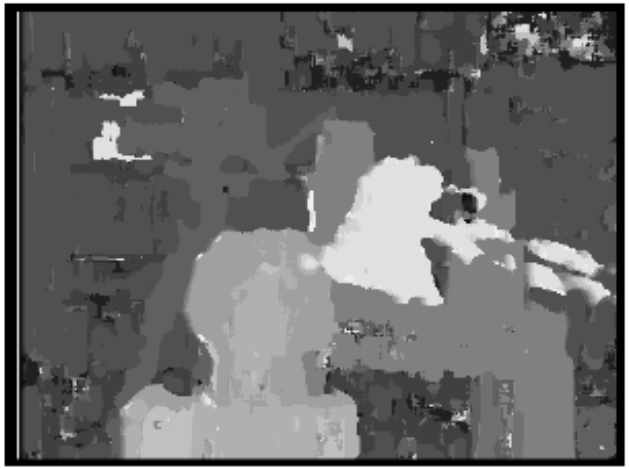
- Compare $m \times m$ patches from two views.
- Form vectors v and v' of length m^2 from them
- Matching scores between patches:
- Sum of Absolute Difference (SAD): $\|v - v'\|_1$
- Sum of squared difference (SSD): $\|v - v'\|_2$





Matching Patches

(AD)



(SAD)

Matching Patches

- Correlation: $\frac{v'^T v}{\sqrt{v^T v}\sqrt{v'^T v'}}$

- Normalized correlation: $\frac{\bar{v}'^T \bar{v}}{\sqrt{\bar{v}^T \bar{v}}\sqrt{\bar{v}'^T \bar{v}'}}$. Range: $[-1, 1]$.

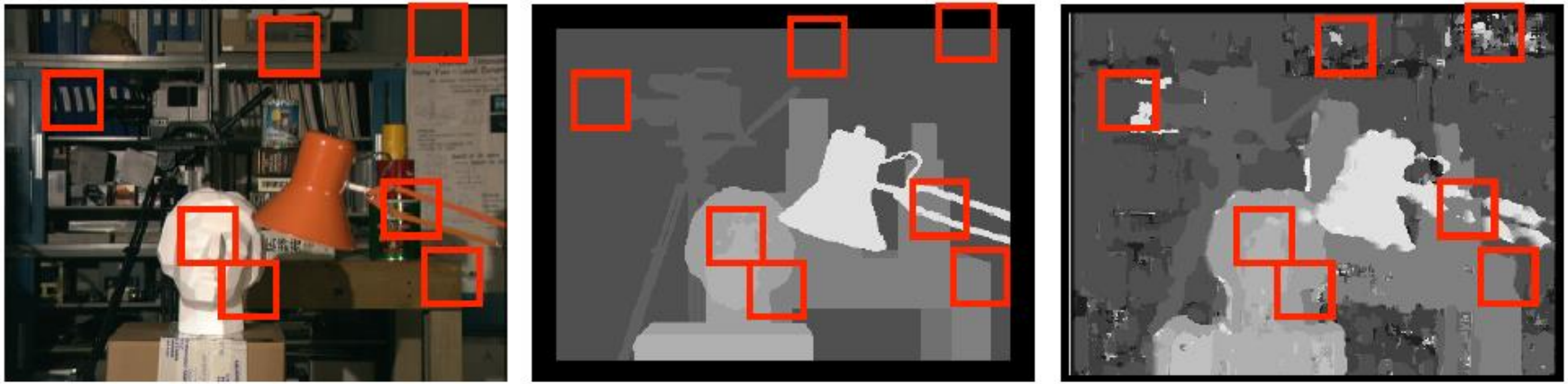
  where $\bar{v}$ and $\bar{v}'$ are vectors with respective patch-mean colour subtracted.

- Invariant to affine changes in intensity/colour.

Matching Patches

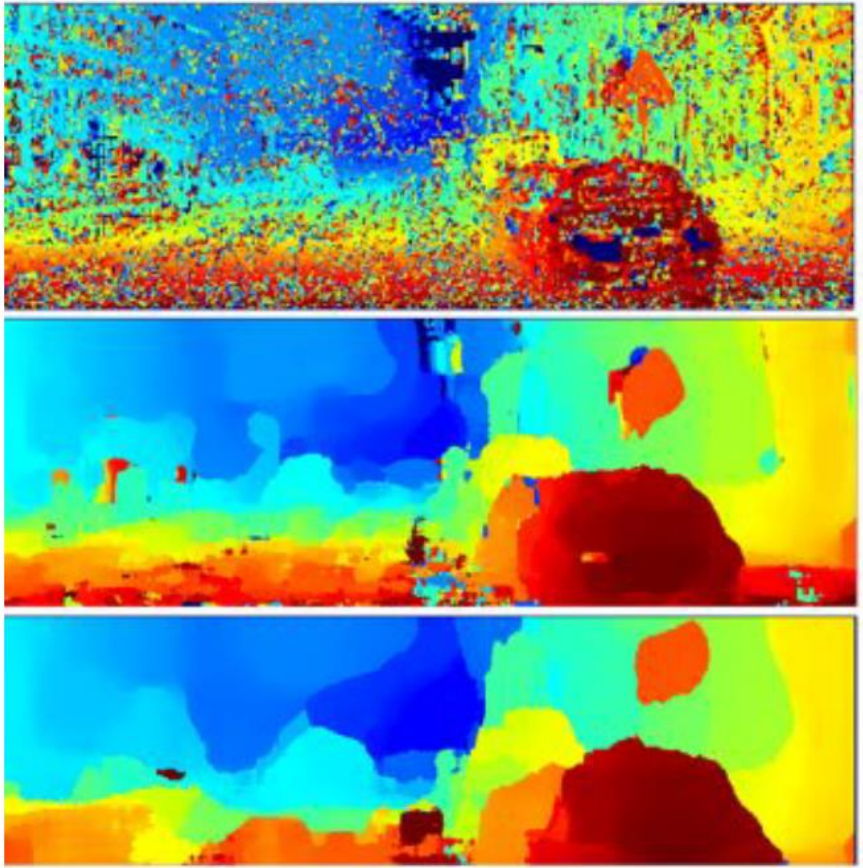
- Issues with patch-based approach
  - Assumes constant depth within the patch
    - Invalid for depth discontinuities
    - Invalid for slanted/non-planar surfaces

— Repetitive textures
— Uniform areas
— Thin structures



Matching Patches

- Varying Patch Sizes



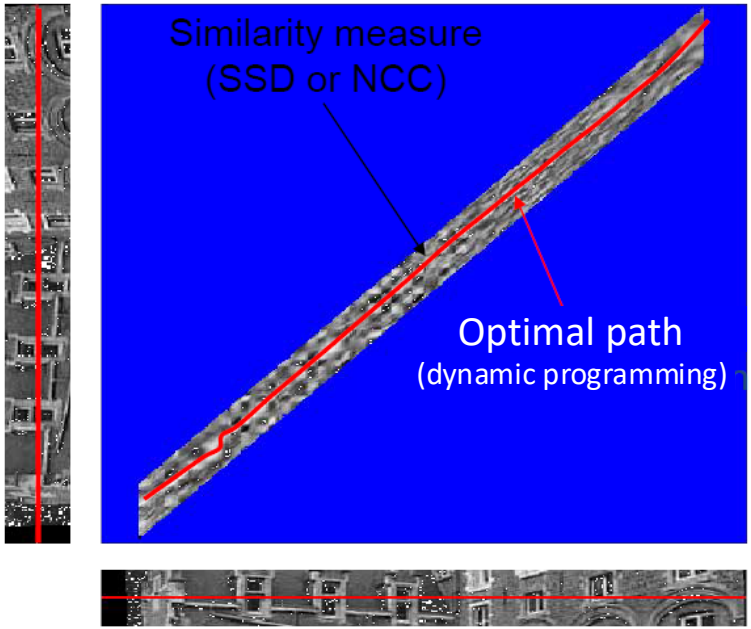
patch size = 5

patch size = 35

patch size = 85

Global Matching with Dynamic Programming

- Improving correspondence: Non-local constraints



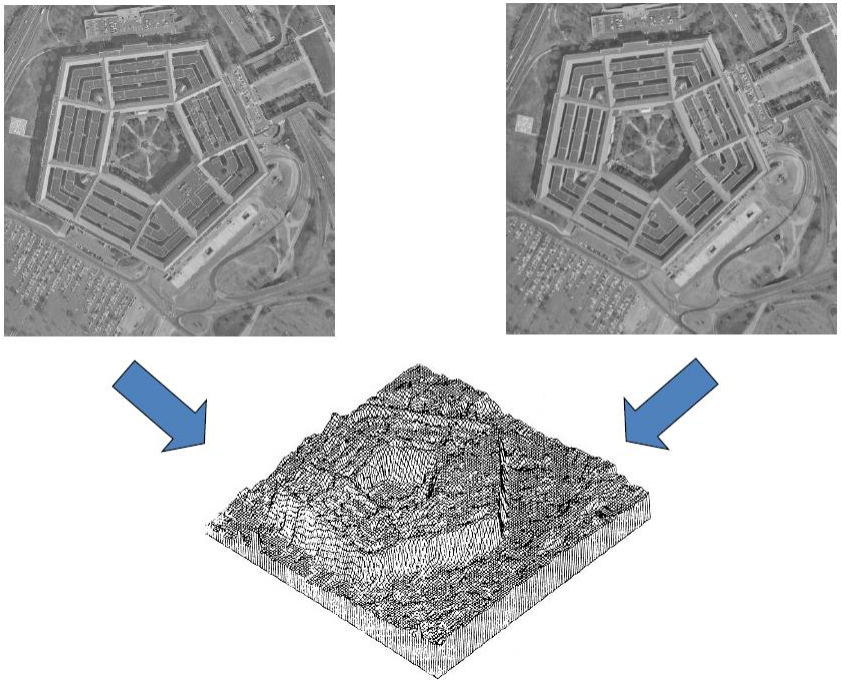
Constraints
- epipolar
- ordering
- uniqueness
- disparity limit

Trade-off
- Matching cost (data)
- Discontinuities (prior)

Consider all path satisfying the constraint and pick the best with dynamic programming

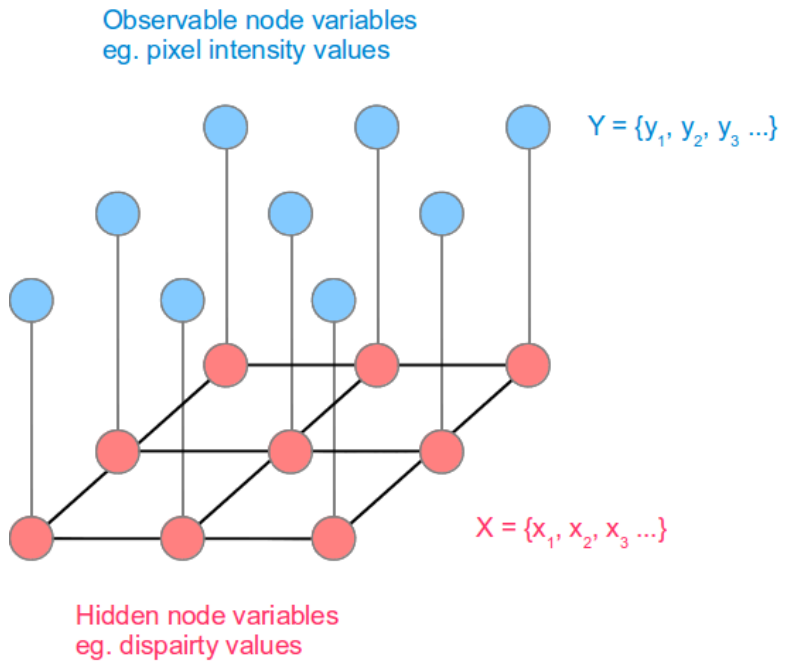
Global Matching with Dynamic Programming

- Ohta and Kanade, 1985



Global Matching with Energy Minimization

- Improving correspondence: Global Optimization
  - With Energy Minimization on top, e.g., a Markov Random Field (MRF)

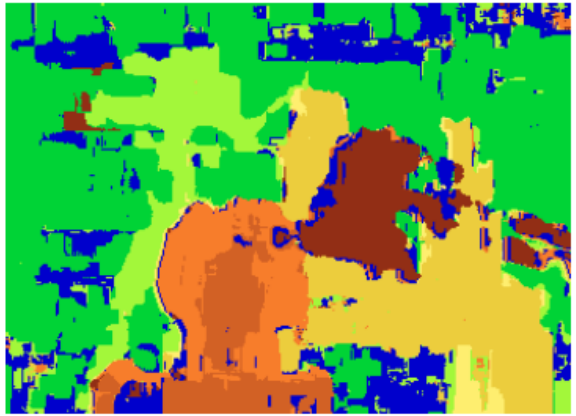


$$E(d) = E_{data}(d) + \lambda E_{smooth}(d)$$

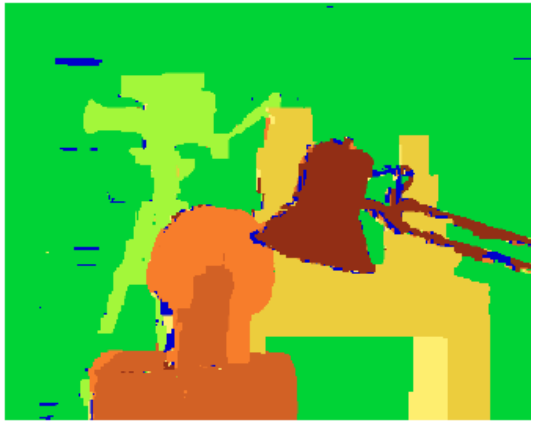
Global Matching with Energy Minimization

- Graph Cut energy minimization

Ground Truth



Local Patch Matching



Graph Cut

Y. Boykov, O. Veksler, and R. Zabih, Fast Approximate Energy Minimization via Graph Cuts, PAMI 2001

Thank You